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SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1715

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2 **Duration: 30 minutes**

Total Marks: 20

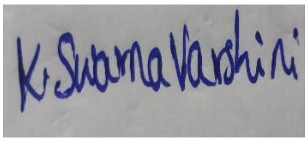
Annexure A

Descriptive Written Test

Code of Conduct:

I _K. Swarna Varshini__(192324081)______________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.

Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Criteria	**Weightage**	**Q1 (10)**	**Q2 (10)**	**Total Marks (20)**
Understanding of Concepts	25% (2.5)	/2.5	/2.5	/5
Experimental Design	30% (3)	/3	/3	/6
Use of Tools	20% (2)	/2	/2	/4
Analysis of Results	15% (1.5)	/1.5	/1.5	/3
Documentation	10% (1)	/1	/1	/2
Total per Question	**10 Marks**	**/10**	**/10**	**/20**

100

ANSWERS

1. Smart Home Automation System – Intelligent Agents

Problem Statement

A smart home automation system controls **lighting, temperature, and security** using real-time inputs such as motion sensors, temperature sensors, and time of day. The system should also learn user preferences to improve comfort, energy efficiency, and safety.

Types of Intelligent Agents

1. Simple Reflex Agent

Working:

- Acts only on the current input using predefined rules.
- Does not remember previous events.

Example in Smart Home:

- If motion is detected → Turn ON lights.
- If no motion for 10 minutes → Turn OFF lights.
- If door opens unexpectedly at night → Trigger alarm.

Advantages

- Fast response.
- Easy to implement.
- Low computational cost.

Disadvantages

- Cannot learn user habits.
- Cannot predict future situations.
- Works only for simple environments.

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2. Model-Based Reflex Agent

Working:

- Maintains an internal model of the environment.
- Uses current sensor data and previous states for decision-making.

Example

- Remembers whether lights are already ON.
- Detects whether someone is inside the room.

- Keeps track of window status and room occupancy.

Advantages

- Handles partially observable environments.
- Makes better decisions than simple reflex agents.
- Improves energy efficiency.

Disadvantages

- Requires additional memory.
- More complex than simple reflex agents.

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3. Goal-Based Agent

Working:

- Chooses actions that help achieve a specific goal.

Example Goals

- Maintain room temperature at **24°C**.
- Secure the home when everyone leaves.
- Reduce electricity consumption.

Advantages

- Makes intelligent decisions.
- Can plan multiple actions.
- Flexible in changing situations.

Disadvantages

- Planning requires more computation.
- Slower than reflex agents.

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4. Utility-Based Agent

Working:

- Selects the action with the highest utility (benefit).
- Balances comfort, security, and energy efficiency.

Example
If:

- Outside temperature is very high,
- Electricity price is high,
- User prefers 24°C,

The system decides the best AC setting to maximize comfort while minimizing electricity cost.

Advantages

- Produces optimal decisions.
- Handles conflicting objectives.
- Learns user preferences over time.

Disadvantages

- Utility calculation is complex.
- Requires more processing power.

Comparison of Intelligent Agents

Agent Type	Memory	Goal-Oriented	Learns Preferences	Suitable for Smart Home
Simple Reflex	No	No	No	Basic automation
Model-Based	Yes	No	Limited	Better environment tracking
Goal-Based	Yes	Yes	Limited	Planning tasks
Utility-Based	Yes	Yes	Yes	Best overall performance

Most Effective Approach

A Hybrid Agent combining:

- Simple Reflex Agent for immediate responses (turning lights ON/OFF).
- Model-Based Agent for maintaining home status.
- Goal-Based Agent for achieving comfort and security goals.
- Utility-Based Agent for balancing comfort, security, and energy efficiency.

Justification

- Immediate reactions ensure quick response.
- Internal memory improves decision-making.
- Goal planning maintains desired conditions.
- Utility optimization minimizes energy consumption while maximizing user comfort and safety.

Therefore, a Hybrid Utility-Based Intelligent Agent is the most effective solution for a smart home automation system**.**

2. Robot Navigation in an Unknown Maze

Problem Statement

A robot is placed in an unknown maze and must find the exit without prior knowledge of the layout. The robot must explore the maze using search algorithms.

Uniform Cost Search (UCS)

Working

- Expands the node with the lowest cumulative path cost.
- Uses a priority queue.
- Always finds the least-cost path.

Example

If paths have different movement costs:

- Left path = Cost 8
- Right path = Cost 5

The robot chooses the right path because it has the lower total cost.

Advantages

- Finds the optimal solution.
- Works with varying path costs.
- Complete algorithm.

Disadvantages

- High memory usage.
- Slow for large search spaces.

Complexity

- **Time:** $O\left(b^{1+\lfloor C^*/\varepsilon \rfloor}\right)$
- **Space:** $O\left(b^{1+\lfloor C^*/\varepsilon \rfloor}\right)$

where:

- b = branching factor
- C^* = optimal path cost
- ε = minimum step cost

Iterative Deepening Search (IDS)

Working

- Repeatedly performs Depth-Limited Search with increasing depth.
- Searches depth 0, then 1, then 2, and so on until the goal is found.

Example

Depth 0:

Start

Depth 1:

Start

├── A

└── B

Depth 2:

Start

├── A

| ├── C

| └── D

└── B

├── E

└── Exit

The robot finds the exit when the required depth is reached.

Advantages

- Very low memory usage.
- Complete.
- Suitable when solution depth is unknown.

Disadvantages

- Repeats node expansion.
- Slower than BFS due to repeated searches.

Complexity

- **Time:** $O(b^d)$
- **Space:** $O(bd)$

where:

- b = branching factor
- d = depth of solution

Comparison of UCS and IDS

Feature	UCS	IDS
Path Cost	Considers cost	Ignores cost
Optimal	Yes	Yes (equal step cost)
Complete	Yes	Yes
Time Complexity	High	Moderate
Space Complexity	High	Low
Suitable for Unknown Depth	Moderate	Excellent

Relation to Intelligent Agents

Simple Reflex Agent

- Makes decisions based only on current surroundings.
- Cannot remember explored paths.
- May repeatedly visit the same location.

Model-Based Agent

- Maintains an internal map of explored areas.
- Avoids revisiting the same paths.
- Suitable for unknown environments.

Goal-Based Agent

- Plans movements toward reaching the exit.
- Uses search algorithms like UCS or IDS.

Utility-Based Agent

- Chooses paths that minimize:
  - Distance
  - Energy consumption
  - Time
  - Risk

Best Combination

Recommended Agent

Model-Based Goal-Based Agent

Recommended Search Algorithm

- **Uniform Cost Search (UCS)** when movement costs vary.
- **Iterative Deepening Search (IDS)** when all moves have equal cost and memory is limited.

Justification

A **Model-Based Goal-Based Agent** is the best choice because it:

- Builds an internal map of the unknown maze.
- Remembers explored paths.
- Plans efficient routes toward the exit.
- Avoids unnecessary repetitions.

When combined with:

- **UCS**, it guarantees the least-cost path in weighted mazes.
- **IDS**, it provides a memory-efficient solution for unweighted mazes.

Conclusion: A **Model-Based Goal-Based Agent using UCS (for weighted mazes) or IDS (for unweighted mazes)** offers the best balance of optimality, memory efficiency, and intelligent decision-making.